Research Statement

Gabriel Angelini-Knoll

I am an algebraic topologist and my work explores the connections between algebraic topology, number theory, and geometric topology. My research program is divided into two categories: the study of periodicity in algebraic K-theory with applications to number theory, and equivariant Hochschild homology with applications to geometric topology. The common thread in my research is the study of invariants of rings, or more generally ring spectra.

One of the main families of conjectures in the area of periodicity and algebraic K-theory are the Ausoni–Rognes red-shift conjectures, which generalize famous conjectures of Lichtenbaum and Quillen. My work gives quantitative evidence for the red-shift conjectures from several different perspectives. I have done this by developing tools for computing Hochschild homology [AK18], giving new computations of Hochschild homology using these tools [AK17, AK21, GCH21, AAC+22], and building on these computations to give evidence [AK18, AS20, AQ19, AK22] for the Ausoni–Rognes red-shift conjectures in algebraic K-theory. I also have several ongoing projects extending this evidence and computational knowledge [AHW22, ACH22]. These results are sumarized in Section 1.

Equivariant topology has had a renaissance recently due to groundbreaking work of Hill, Hopkins, and Ravenel on the Arf–Kervaire invariant one problem in geometric topology. My work uses Hochschild homology and genuine equivariant homotopy theory to approach questions in geometric topology in a similar vein. My work [AQ21] uses equivariant homotopy and Hochschild homology to give a first step towards computing the stable h-cobordism space of a point. Additionally, my work [AGH21] gives new perspectives on Hochschild homology and Witt vectors of rings with anti-involution. My work in progress [AMP22], gives a combinatorial approach to equivariant Hochschild homology. One goal of this program is to study equivariant knot Floer homology using this theory. In work in progress [ABBK22], we also use synthetic homotopy to study C_3-equivariant homotopy, with the long-term goal of resolving the 3-primary Arf–Kervaire invariant problem. Finally, in work in progress [AQ22] we study a new motivic filtration on Hochschild homology of rings with anti-involution, with the goal of studying the goal of studying the arithmetic of rings and ring spectra with anti-involution. These results are summarized in Section 2.

1 Algebraic K-theory and arithmetic of ring spectra

This section is split into four subsections: Section 1.1 a brief overview of the red-shift conjectures, Section 1.2 a brief overview of trace methods, Section 1.3 a summary of results, and Section 1.4 a summary of work in progress.

1.1 Higher Lichtenbaum–Quillen conjectures

Just as rings are monoids in the category of abelian groups, ring spectra are monoids in the category of spectra. Broadly, the Ausoni–Rognes red-shift conjectures are about the arithmetic of ring spectra. In the late 1960's, Adams [Ada66] and Quillen [Qui72] showed

a v_1-periodic family known as the α-family is detected in $\mathrm{K}_{4k-1}(\mathbb{Z})$. This was the first evidence for a conjecture of Lichtenbaum that certain quotients of orders of algebraic K-theory groups

$$|\mathrm{K}_{4k-2}(\mathcal{O}_F)|/|\mathrm{K}_{4k-1}(\mathcal{O}_F)| = \zeta_F(-2k+1)$$

can be identified (up to a sign and power of two) with special values of the Dedekind zeta function of a number field F with ring of integers $\mathcal{O}_F$. My work [AK18] gives the first indication of a higher chromatic height analogue of this conjecture (see Section 1.3).

Question 1.1. [Higher Lichtenbaum] Is there a precise relation between $\pi_* K(K(\mathcal{O}_F))$ and modular forms over $\mathcal{O}_F$?

The Ausoni–Rognes red-shift conjectures generalize the Lichtenbaum–Quillen conjectures to higher chromatic heights and they remain under active investigation with some significant advances in the last five years. In homotopy theory, there are more objects that behave like prime fields, which I will use to phrase some of the red-shift conjectures. In addition to the finite fields $\mathbb{F}_q$ and and the rationals $\mathbb{Q}$ there ring spectra called Morava K-theory $K(n)$ with coefficients $K(n)_* = \mathbb{F}_p[v_n^{\pm 1}]$. Just as we can rationalize an abelian group, we can localize a spectrum X at Morava K-theory. We denote the localization at the sum $K(0) \oplus \cdots \oplus K(n)$ by L_n.

For the statement of the Ausoni–Rognes red-shift conjecture, we also need to define finite versions L_n^f of the localizations L_n. A finite spectrum $F(n-1)$ has type n if $L_{n-1}F(n-1) = 0$ and $L_n F(n) \neq 0$. In this case, the spectrum $F(n-1)$ has a self map

$$v\colon\ \Sigma^d F(n-1) \longrightarrow F(n-1)$$

and we write $T(n-1) = v^{-1}F(n-1)$ for the colimit of the iterated composite. This allows us to consider a variant of L_n by localizing at the sum $T(0) \oplus \cdots \oplus T(n)$ denoted L_n^f. It was conjectured by Ravenel that the localizations L_n and L_n^f agree, a conjecture called the telescope conjecture, but it is now widely believed that this conjecture is false and it remains a deep open problem.

In the phrasing of Waldhausen [Wal84], the Lichtenbaum–Qullen conjectures state that the map

$$\pi_s \mathrm{K}(R[1/p])_{(p)} \longrightarrow \pi_s L_1^f \mathrm{K}(R[1/p])_{(p)}$$

is an isomorphism for sufficiently large s. In other words, the fiber of the localization map is co-connected. We will use the following terminology: A map $X \to Y$ with co-connected fiber is truncated. The Ausoni–Rognes red-shift conjecture may be stated as follows.

Question 1.2. [Higher Lichtenbaum–Quillen] Suppose R is a ring spectrum and the map $R_{(p)} \longrightarrow L_n^f R_{(p)}$ is truncated. Is the map $\mathrm{K}(R)_{(p)} \longrightarrow L_{n+1}^f \mathrm{K}(R)_{(p)}$ truncated?

This reduces to the Lichtenbaum–Quillen conjecture when $n = 0$. In [AR08], this is only stated for suitable finite extensions of the $K(n)$-local sphere, but this more general version is implied by an earlier version of the conjecture from [Rog00] (See Section 1.5). A weak form of this conjecture has been posed for $\mathbb{E}_2$-rings and it is a step towards Question 1.2.

Question 1.3. [Qualitative Red-shift] For an $\mathbb{E}_2$-ring R, $L_{K(n)}R = 0$ iff $L_{K(n+1)}K(R) = 0$.

The original red-shift problem posed by Rognes in [Rog00] uses the notion of pure fp-type based on work of [MR99] and TC, which closely approximates algebraic K-theory (see Section 1.2).

Definition 1.4. A spectrum X has pure fp-type n if $F(n-1)_*X$ is a finitely generated free $P(v)$-module. A spectrum is fp-type n if $F(n)_*X$ is finite.

Question 1.5. If R has pure fp-type n, then does $\mathrm{TC}(R)$ has pure fp-type $n+1$?

An afffirmative answer to Question 1.5 implies an affirmative answer to Question 1.2. Also, an affirmative answer to Question 1.2 implies an affirmative answer to Question 1.3. However, the reverse implications do not hold. In my work and my work in progress I give evidence for each of these questions, summarized in Sections 1.3–1.4.

1.2 Trace methods

My approach to algebraic K-theory of a connective ring spectrum R uses the theory of trace methods developed by [BHM93], where one computes algebraic K-theory by successive approximations. The first approximation to algebraic K-theory is Hochschild homology. Classically, one uses genuine equivariant homotopy theory and genuine cyclotomic structure to construct $\mathrm{TR}(R)$ from Hochschild homology and then define p-complete TC as the homotopy equalizer

$$\mathrm{TC}(R)^\wedge_p = \mathrm{eq}\left(\mathrm{TR}(R) \underset{F}{\overset{\mathrm{id}}{\rightrightarrows}} \mathrm{TR}(R) \right).$$

The groups $\mathrm{TR}_*(R)$ are interesting in their own right because they recover the de Rham Witt complex of a commutative ring [HM03].

Groundbreaking work of Nikolaus–Scholze [NS18] simplified the computational approach to topological cyclic homology $\mathrm{TC}(R)$ by describing TC as the homotopy equalizer

$$\mathrm{TC}(R) = \mathrm{eq}\left(\mathrm{TC}^-(R) \underset{\varphi}{\overset{can}{\rightrightarrows}} \mathrm{TP}(R)^\wedge \right)$$

of two maps where $(-)^\wedge$ denotes pro-finite completion. The topological negative cyclic homology $\mathrm{TC}^-_*(R)$ and topological periodic cyclic homology $\mathrm{TP}_*(R)$ are topological analogues of (algebraic) negative cyclic homology and periodic cyclic homology and they are easier to calculate because they do not rely on genuine equivariant homotopy theory. The intermediate approximations $\mathrm{TC}^-(R)$ are $\mathrm{TP}(R)$ to algebraic K-theory theory are interesting in their own right because of the relation to prismatic cohomology [BS19].

Finally, Dundas–Goodwillie–McCarthy [DGM13] proved that topological cyclic homology is a close approximation to algebraic K-theory in the sense that if $f\colon A \to B$ is a map of ring spectra such that $\pi_0 f$ is surjective with nilpotent kernel and $E(A,B)$ is the fiber of $E(f)$ for $E \in \{K, \mathrm{TC}\}$, we have an equivalence $\mathrm{K}(A,B) \simeq \mathrm{TC}(A,B)$. This gives an explicit computational approach to algebraic K-theory, which has led to numerous advances in the subject, for example [HM03, AR02, HW20, AAC+22].

1.3 Results on higher Lichtenbaum–Quillen conjectures

One of my primary research goals has been to study the arithmetic of ring spectra. I have approached this by providing evidence for the Ausoni–Rognes red-shift conjectures (see Section 1).

There are v_n-periodic families of elements in the p-local stable homotopy groups of spheres known as the p-primary n-th Greek letter families. When $p \geq 3$ and $n = 1$, the α-family has a close relationship to special values of the Riemann zeta function by work of Adams [Ada66] and this family of elements is detected in the algebraic K-theory of the integers by Quillen [Qui72]. These results were one of the results that motivated Lichtenbaum [Lic73] to speculate that algebraic K-theory groups and special values of zeta functions should be closely related more generally. In [Lau99, Beh06, BL09], Behrens and Laures demonstrated a tight connection between the p-primary β-family for $p \geq 5$ and integral modular forms. The following result is therefore a higher chromatic height analogue of the results of Adams and Quillen and gives evidence for Question 1.1.

Theorem 1.6 ([AK18])**.** The p-primary β-family modulo (p, v_1) is detected in the iterated algebraic K-theory $\pi_* \mathrm{K}(\mathrm{K}(\mathbb{Z}))$ of the integers for $p \geq 5$.

This result built on computations in my PhD thesis [AK17], published in [AK21] and joint work with Salch [AS18]. In [AS18], we provided a useful tool for computing (generalizations of) Hochschild homology of commutative ring spectra. For simplicity, we state the result only for Hochshild homology (THH). By a multiplicatively filtered commutative ring spectrum, I mean a sequence of spectra $\longrightarrow I_2 \longrightarrow I_1 \longrightarrow I_0$ equipped with structure maps $\rho_{i,j} \colon I_i \wedge I_j \longrightarrow I_{i+j}$ for all $i, j \geq 0$ satisfying certain coherence diagrams. There is an associated graded functor E_0^* from multiplicatively filtered commutative ring spectra to commutative ring spectra.

Theorem 1.7 ([AS18])**.** Let I be a multiplicatively filtered commutative ring spectrum, then there is a spectral sequence

$$E_1 = E_* \left(\mathrm{THH}(E_0^* I)\right) \implies E_* \left(\mathrm{THH}(I_0)\right)$$

for any generalized homology theory E.

In [AK21], I used this to compute mod (p, v_1) Hochschild homology of the connective cover of the $K(1)$-local sphere, denoted simply j, at primes $p \geq 5$.

Theorem 1.8 ([AK21])**.** Let $p \geq 5$. There is an isomorphism of graded $\mathbb{F}_p$-algebras

$$V(1)_* \mathrm{THH}(j) \cong H(E(\alpha_1, \lambda_1, \lambda_2); d) \otimes P(\mu_2) \otimes \Gamma(\sigma b)$$

where $d(\lambda_2) = \alpha_1 \lambda_1'$.

Towards answering Question 1.3, I proved several $K(n)$-local vanishing results. There are ring spectra $y(n)$ that filter between $H\mathbb{F}_2$ and $\mathbb{S}$

$$\mathbb{S} = y(0) \longrightarrow y(1) \longrightarrow \cdots \longrightarrow y(\infty) = H\mathbb{F}_2.$$

We say $R \to S$ is a $K(n)$-local equivalence if the fiber F satisfies $K(n)_* F = 0$.

Theorem 1.9 ([AQ19])**.** Whenever the map $y(n) \to H\mathbb{F}_2$ is a $K(m)$-local equivalence, the map $\mathrm{K}(y(n)) \longrightarrow \mathrm{K}(\mathbb{F}_2)$ is a $K(m)$-local equivalence and the map $\mathrm{TP}(y(n)) \longrightarrow \mathrm{TP}(\mathbb{F}_2)$ is a $K(m+1)$-local equivalence.

This shows that algebraic K-theory does not decrease chromatic complexity and TP increases chromatic complexity for the family of ring spectra $y(n)$. This is interesting (and challenging) in particular because $y(n)$ is a ring spectrum, but is not an $\mathbb{E}_2$-ring so algebraic K-theory of $y(n)$ doesn't have any additional ring structure. The result uses a technical result about commuting Morava K-theory with certain sequential limits proven in joint work with Salch [AS20]. As an application of this technical result, we also prove an upper Morava K-theory vanishing region for commutative ring spectrum forms of $BP\langle n\rangle$. Note that $L_{K(m)}BP\langle n\rangle = 0$ for $m \geq n+1$.

Theorem 1.10 ([AS20])**.** When B is a commutative ring spectrum form of $BP\langle n\rangle$, then $L_{K(m)}K(B) = 0$ for $m \geq n+2$.

The calculations of $\mathrm{THH}_*(BP\langle n\rangle)$ for $n \geq -1$ are deep results and they are increasingly difficult as n increases. For example, Bökstedt's result $\mathrm{THH}_*(BP\langle -1\rangle) = P(\mu_0)$ [Bök87] can be used to prove Bott periodicity [HN19]. When describing the torsion in $\mathrm{THH}_*(BP\langle 1\rangle)$, Angeltveit–Hill–Lawson say it "follows a kind of tower-of-Hanoi pattern with increasingly complicated, inductively built, components" [AHL10, p.6]. In future work, I plan to give a complete description of $\mathrm{THH}_*(BP\langle 2\rangle)$. Towards this aim, we have already proven.

Theorem 1.11 ([GCH21])**.** When $m = 0, 1, 2$, there are isomorphisms

$$\mathrm{THH}_*(BP\langle 2\rangle; k(m)) \cong P(v_m) \otimes E(\lambda_1, \ldots, \lambda_{2-m}) \otimes T^n_m,$$

where T^n_m is an explicit v_m-torsion module.

In work in progress joint with D. Culver and E. Höning, I use different filtration techniques such as Theorem 1.7 and Bockstein spectral sequences to compute $\mathrm{THH}_*(BP\langle 2\rangle; \ell)$ in [ACH22].

To answer Question 1.5, we wish to give explicit computations of $F(n-1)_*K(BP\langle n\rangle)$ when $n = 0$. Let $V(n-1) = \mathbb{S}/(p, \ldots, v_{n-1})$ be a Smith–Toda complex. Bökstedt–Madsen [BM94] computed $V(0)_* \mathrm{TC}(\mathbb{Z}_p)$ and consequently $V(0)_* \mathrm{K}(\mathbb{Z}_p)$. Ausoni–Rognes [AR02] computed $V(1)_* \mathrm{TC}(BP\langle 1\rangle)$ and consequently $V(1)_* \mathrm{K}(BP\langle 1\rangle)$, which led Rognes to pose Question 1.5. In joint work [AAC+22] with C. Ausoni, D. Culver, E. Höning, and J. Rognes I compute $V(2)_* \mathrm{TC}(BP\langle 2\rangle)$ and $V(2)_* \mathrm{K}(BP\langle 2\rangle)$ at primes $p \geq 7$ improving on a result from the landmark paper of Hahn–Wilson [HW20] in the case $n = 2$. This is also currently the first and only known result proving the strong form of the Ausoni–Rognes red-shift conjecture at higher heights. Note that $BP\langle 2\rangle$ has pure fp-type 2.

Theorem 1.12 ([AAC+22])**.** The spectrum $\mathrm{TC}(BP\langle 2)$ has pure fp-type 3.

1.4 Work in progress on the arithmetic of ring spectra

Currently, I am working on extending the current quantitative knowledge of the Ausoni–Rognes red-shift conjectures and more generally the arithmetic of ring spectra. Work of

Ausoni–Rognes [AR12] suggests that the cofiber of the transfer map sitting in the cofiber sequence

$$K(L/p) \to K(L_p^\wedge) \to K(ffL_p)$$

behaves like the algebraic K-theory of the fraction field of L_p in analogy with the classical localization sequence

$$K(\mathbb{F}_p) \to K(\mathbb{Z}_p^\wedge) \to K(\mathbb{Q}_p)$$

due to Quillen. In order to define such a sequence one must leave the algebraic K-theory of rings and pass to the algebraic K-theory of rings with many objects. One must also make sense of G-theory for ring spectra and ring spectra with many objects. Making these notions precise is the subject of ongoing joint work with C. Ausoni, D. Culver, E. Höning, and J. Rognes.

Recent work of Hahn–Raksit–Wilson [HRW22] make computations of algebraic K-theory and topological cyclic homology more accessible. In [HRW22], the authors provide a motivic filtration on topological cyclic homology of commutative ring spectra, which is amenable to computation for certain ring spectra called chromatically quasi-syntomic ring spectra. As an application they extend computations of $V(1)_* \operatorname{TC}(BP\langle 1\rangle)$ to $p = 3$ and provide a much simpler approach to the computation in this case. Even though ring spectra such as the truncated Brown–Peterson spectra $BP\langle n\rangle$, the connective Morava K-theory $k(n)$, and the image of J-spectrum j are not chromatically quasisyntomic, they may be approachable by similar means and this is the subject of ongoing work with subsets of my collaborators J. Hahn, D. Wilson, C. Ausoni, E. Höning, and J. Rognes.

For chromatically quasisyntomic rings, the motivic filtrations on TC and TP due to [HRW22] extend the motivic filtration on TC and TP by [BMS19, BS19, BL22], whose associated graded are syntomic cohomology and (absolute) prismatic cohomology, respectively. One may therefore define syntomic cohomology and prismatic cohomology for ring spectra using this approach. I will take this as a definition of syntomic cohomology and prismatic cohomology for an arbitrary ring spectrum. I will now enumerate some goals that I think are within reach with these methods in the near future.

Goal 1.13. Compute the syntomic cohomology of truncated Brown–Peterson spectra $BP\langle n\rangle$, connective Morava K-theory $k(n)$, and the image of J spectrum j.

The syntomic cohomology is the E_2-page of a spectral sequence computing $\operatorname{TC}(R)$ and it often collapses at this page for bidegree reasons. For example, this is the case for $V(1)_*K(BP\langle 1\rangle)$ by [HRW22]. In general, there may be longer differentials in this spectral sequence. Nevertheless, there are questions that you can answer using this approach such as Question 1.2. We say X has fp-type n if $F(n)_*X$ is finite for some type $n+1$-spectrum $F(n)$. For example, $k(n)$ has fp-type n, j has fp-type 1, and ko has fp-type 1.

Goal 1.14. Prove that $K(k(n))$ has fp-type $n+1$ and $K(ko)$.

This implies an affirmative answer to Question 1.2 in these cases. In general, I believe a regularity hypothesis is necessary for K-theory of an fp-type n ring spectrum to be fp-type $n+1$. For example, I expect that $K(j)$ is not fp-type 2.

Goal 1.15. Determine whether $K(j)$ has fp-type 2.

The telescope conjecture states that $L_n = L_n^f$ and it is now widely believed to be false. In [MR99], they conjecture that the telescope conjecture is true for all fp-spectra, so if one believes this conjecture than an spectrum that is not fp (does not have finiltely presented mod p cohomology as a module over the Steenrod algebra) is a good choice for studying the red-shift conjecture. Explicit computations also suggest that $V(1)_*K(j))$ may be a good test case for the telescope conjecture, as has been suggested by J. Hahn. I will say that the telescope conjecture holds for X if $L_nX = L_n^fX$.

Goal 1.16. Determine whether the telescope conjecture holds for $V(1) \otimes K(j)$.

2 Equivariant topology and geometric applications

This section is split into four sub-sections: Section 2.1 a brief survey of equivariant topology, Section 2.2 a brief survey of motivic topology, Section 2.3 on equivariant Hochschild homology, and Section 2.4 on interactions between equivariant topology and motivic topology

2.1 Equivariant algebra and topology

The field of equivariant topology was revitalized in the last decade with the groundbreaking work of Hill, Hopkins, and Ravenel [HHR16] resolving almost all remaining cases of the Arf–Kervaire invariant one problem. Since this work, there has been significant advances in the subject building on the machinery in their paper. One of the foundational parts of their work is the norm functor,

$$N_H^G \colon \mathsf{Sp}^H \longrightarrow \mathsf{Sp}^G,$$

which they define for any finite group G.

The work of [HHR16] also relied on the theory of genuine commutative G ring spectrum. One may ask what the associative analogue of a genuine commutative G-ring spectrum is and one answer, in the case when G is the cyclic group of order 2, is a ring spectrum with anti-involution. Ring spectra with anti-involution play a key role in surgery theory as they are the input for Real algebraic K-theory, a genuine equivariant version of algebraic K-theory, which recovers various forms of L-theory. Just as classically algebraic K-theory can be approximated by topological Hochschild homology, Real algebraic K-theory can be approximated by Real topological Hochschild homology. One may then form Real analogues of topological cyclic homology and related invariants and use them to approximate Real algebraic K-theory.

My work in equivariant homotopy theory falls into four projects: the study of norms for the group $O(2)$ of two-by two orthogonal matrices and equivariant Witt vectors, generalizations of real topological cyclic homology to other compact Lie groups, new perspectives of equivariant homotopy theory for odd order cyclic groups, and the study of the strongly even filtration on THR.

2.2 Motivic topology

In motivic topology, one replaces topological spaces with schemes and then does homotopy theory in this setting. The role of the interval, which is contractible classically, is played by

the affine line $\mathbb{A}^1$. One therefore considers pre-sheaves on the category of smooth schemes over a base scheme $X = \mathrm{spec}(R)$ for some commutative ring R, localizes in the Nisnevich topology, and then inverts $\mathbb{A}^1$ so that the category behaves more like the category of topological spaces. Motivically, the scheme $\mathbb{G}_m$ behaves like a second sphere and smashing this with the usual sphere produces projective space $\mathbb{P}^1$. The analogue of infinite loops spaces are then infinite loop spaces with respect to the sphere $\mathbb{P}^1$ and stabilizing with respect to $\mathbb{P}^1$ produces the stable category $\mathrm{SH}(R)$.

The motivic stable category $\mathrm{SH}(\mathbb{C})$ has a realization functor to the category of spectra Sp called Beti realization and the extra grading on the homotopy groups in $\mathrm{SH}(\mathbb{C})$ coming from the additional spheres $\mathbb{G}_m^{\otimes t}$ has led to significant progress on computations, such as the homotopy groups of spheres by Isaksen–Wang–Xu [IWX20]. In C_2-equivariant homotopy, their is a similar Beti realization functor from $\mathrm{SH}(\mathbb{R})$ and C_2-spectra, which simplifies C_2-equivariant homotopy theory computations.

2.3 Equivariant Hochschild homology

In [ABG+18], they define norms $N_{\mu_n}^{\mathbb{T}}(R)$ from the groups of unity to the circle group $\mathbb{T}$. In [BGHL18], the authors then used this theory to define a new theory of Witt vectors and Hochschild homology for associative μ_n-Green functors.

In joint work with Gerhardt and Hill [AGH21], I define the norm $N_{D_2}^{O(2)}(R)$ for a ring spectrum with anti-involution R, where D_2 is the cyclic group of order 2. To prove that our definition is reasonable, we prove a fundamental property of the norm functor.

Theorem 2.1 ([AGH21])**.** The restriction of the functor $N_{D_2}^{O(2)}$ to the category of genuine commutative D_2-spectra

$$N_{D_2}^{O(2)}\colon \mathrm{CAlg}(\mathsf{Sp}^{D_2}) \longrightarrow \mathrm{CAlg}(\mathsf{Sp}^{O(2)})$$

is left adjoint to the restriction functor $\iota_{D_2}^*$.

Another property of the norm for finite groups is the multiplicative double coset formula. For our new notion of norm $N_{D_2}^{O(2)}$, we also prove a such a formula, extending [DMPP17].

Theorem 2.2 ([AGH21])**.** Under cofibrancy hypotheses on R, there is an equivalence of D_{2m}-spectra

$$\iota_{D_{2m}}^* N_{D_2}^{O(2)}(R) \simeq N_{D_2}^{D_{2m}} R \wedge^{\mathbb{L}}_{N_e^{D_{2m}} \iota_e^* R} N_{\triangle}^{D_{2m}} \iota_{\triangle}^* c_\zeta R$$

where ζ is the generator of $\mu_{2m} \subset D_{4m}$ and $\triangle = \zeta D_2 \zeta^{-1}$.

We use these results to motivate a new algebraic definition of Real Hochschild homology $\underline{\mathrm{HR}}_*^{D_{2m}}(\underline{M})$ of a ring with anti-involution $\underline{M}$. We prove that this construction is equipped with structure maps

$$R\colon \underline{\mathrm{HR}}_0^{D_{2p^k}}(\underline{M})^{\mu_{p^k}} \longrightarrow \underline{\mathrm{HR}}_0^{D_{2p^{k-1}}}(\underline{M})^{\mu_{p^{k-1}}}.$$

as well as Frobenius and Verschiebung maps F and V coming from the equivariant structure. Therefore, we define the p-typical Real Witt vectors

$$\underline{\mathbb{W}}(\underline{M}; p) = \lim_R \underline{\mathrm{HR}}_0^{D_{2p^k}}(\underline{M})^{\mu_{p^k}}$$

and prove the following refinement of a result of Hesselholt [Hes97], where we write TCR for Real topological cyclic homology.

Theorem 2.3 ([AGH21])**.** Given a ring spectrum R with anti-involution such that a certain $R^1 \lim$ vanishes

$$\mathrm{TCR}_{-1}(R) = \mathbb{W}(\underline{\pi}_0^{D_2} R; p)_F$$

where $\mathbb{W}(\underline{M}; p)_F$ is the coinvariants of an operator F on $\mathbb{W}(\underline{M}; p)$.

In joint work with Péroux and Merling [AMP22], we extend the theory of norms to more general compact Lie groups. To do this, we use the theory of crossed simplicial groups. Crossed simplicial groups $\mathbf{\Delta G}$ are a common generalization of Connes' cyclic category Λ and simplicial groups. Given a crossed simplicial group $\mathbf{\Delta G}$, we construct a notion of Hochschild homology $\mathrm{TH}\mathbf{G}$ whose input is a twisted G_0-ring, where G_0 is a group associated to the crossed simplicial group $\mathbf{\Delta G}$.

A twisted G-ring associated to a group with a group homomorphism $\varphi\colon G \to C_2 = \{-1, 1\}$ is a ring with a G-action by abelian group homomorphisms where g acts by a ring homomorphism if $\varphi(g) = 1$ and a ring anti-homomorphism if $\varphi(g) = -1$. In the study of twisted G-rings, we prove the following result, which we state as a pre-theorem since it is work in progress.

Pre-theorem 2.1. The ∞-category $N\mathbf{\Delta G}_+$ is the ∞-operad for twisted G-rings.

Using this, we give a Nikolaus–Scholze type construction of

$$N^{\mathbf{G}}_{\mathfrak{S}_0}(R) = \mathrm{TH}\mathbf{G}(R)$$

providing new combinatorial models for norm functors for certain compact Lie groups.

Of particular interest is the case of the quaternioic category $\mathbf{\Delta Q}$ whose associated Hochschild homology gives a construction of the norm

$$N^{\mathrm{Pin}(2)}_{C_4}(R).$$

We aim to use $N^{\mathrm{Pin}(2)}_{C_4}(R)$ to study involutive knot Floer homology in analogy with the relationship between Hochschild homology and knot Floer homology. As an example of an application we have the following goal.

Long-Term Goal 2.4. Construct a Hodge-to-de Rham spectral sequence for $N^{\mathrm{Pin}(2)}_{C_4}(R)$ that can be used to study knot Floer homology.

As an example computation, we compute $\mathrm{TH}\mathbf{G}(R[M])$ where R is a twisted G_0-ring and M is a twisted G_0-monoid, which we describe qualitatively here.

Pre-theorem 2.2. The assembly map for $\mathrm{TH}\mathbf{G}$ of the twisted G_0-ring $R[M]$ is an equivalence.

As a completely distinct question, this new theory has potential to receive a natural trace map from equivariant algebraic K-theory.

Long-Term Goal 2.5. Construct a trace map

$$\mathrm{K}_G(R) \longrightarrow N_G^{G\times\mathbb{T}}(R).$$

To give a further refinement of the trace map, we aim to extend the theory of cyclotomic spectra for any crossed simplicial group equipped with an interstice duality $D\colon \mathbf{\Delta G}^{\mathrm{op}} \to \mathbf{\Delta G}$, which is a functor satsifying $D \circ D^{op} = \mathrm{id}$. For example, a large portion of these categories were classified by [DK14] as the planar crossed simplicial groups corresponding to certain coverings of $O(2)$. Long term, this would allow one to compute G equivariant analogues of algebraic K-theory, Real algebraic K-theory, and quaternionic algebraic K-theory using techniques similar to work in progress of Harpaz–Nikolaus–Shah [HNS22] in the case of Real algebraic K-theory.

2.4 Equivariant topology and motivic topology

As discussed in Section 2.2, we can use motivic homotopy theory $\mathrm{SH}(\mathbb{R})$ over $\mathrm{spec}(\mathbb{R})$ to simplify computations in Sp^{C_2}. In the case of a cyclic group of order p, denoted C_p, where p is odd there is not a known good replacement for $\mathrm{SH}(\mathbb{R})$. In work in progress with Behrens, Belmont, and Kong [ABBK22], I aim to give a purely synthetic construction equipped with a realization functor to Sp^{C_p}.

Synthetic homotopy theory was initiated by [GIKR18] and fully developed in [Pst18]. Recently, this was used by Burkland–Hahn–Senger [BHS20] to extend the cellular $\mathbb{R}$-motivic stable homotopy category in a way that allows for realization functors recovering more information about the homotopy groups of C_2-spectra. This extension of the cellular $\mathbb{R}$-motivic stable homotopy category is called the Artin–Tate $\mathbb{R}$-motivic category.

Short-Term Goal 2.6. Give a description of the a-complete Artin–Tate $\mathbb{R}$-motivic category $\mathrm{SH}^{AT}(\mathbb{R})^\wedge_a$ that extends from C_2 to C_p for an odd prime p.

We currently have such a presentation, but since this is work in progress I will not spell out the details here. To extend this to a description of the full synthetic analogue of the $\mathbb{R}$-motivic category one would need interpretations of the other terms in the associated fracture square that do not involve $MU_{\mathbb{R}}$.

Ravenel has suggested that a nice C_p-equivariant Brown–Peterson spectrum BP could be used to resolve the remaining case of the odd primary Arf–Kervaire invariant one problem in analogy with the solution of the 2-primary Kervaire invariant one problem by [HHR16]. The key part of this being the existence of a nice Adams–Novikov spectral sequence based at this C_p-equivariant BP. Some progress in this direction appears in work of Hahn–Senger–Wilson [HSW20]. In our work, we skip the construction of BP and instead aim to construct an analogous spectral sequence.

Long-Term Goal 2.7. Resolve the 3-primary Arf–Kervaire invariant one problem using our new artificial 3-primary analogue of $\mathrm{SH}(\mathbb{R})$.

In a completely different vein, one may attempt to carry out a thorough study of the arithmetic of equivariant ring spectra in analogy with the computations in Section 1.3 and Section 1.4. For rings, there is still a lot that isn't known in this context, for example one may want to define C_2-equivariant prismatic cohomology and syntomic cohomology. In joint

work with J.D. Quigley, we propose to use the strong even filtration, a version of the even filtration of [HRW22] adapted to this context. We will call this filtration on Real topological Hochschild homology the motivic filtration and the spectral sequence associated to this filtration the motivic spectral sequence. In the non-equivariant setting, these filtrations do recover motivic filtrations that were studied previously.

Short-Term Goal 2.8. Compute $\mathrm{THR}(BP_{\mathbb{R}}\langle 1\rangle)$ using descent along the motivic spectral sequence.

Long term, we sim to understand syntomic/prismatic cohomology for suitable commutative monoids in C_2-spectra (also known as N_∞-rings).

Long-Term Goal 2.9. Develop the theory of C_2-equivariant chromatic quasisyntomic N_∞-rings and the notions of syntomic/prismatic cohomology in this context.

References

[AAC+22] Gabriel Angelini-Knoll, Christian Ausoni, Dominic Leon Culver, Eva Höning, and John Rognes. Algebraic K-theory of elliptic cohomology. *arXiv e-prints*, page arXiv:2204.05890, April 2022.

[ABBK22] Gabriel Angelini-Knoll, Mark Behrens, Eva Belmont, and Hana Kong. The modified adams spectral sequence and motivic homotopy. *In progress*, 2022.

[ABG+18] Vigleik Angeltveit, Andrew J. Blumberg, Teena Gerhardt, Michael A. Hill, Tyler Lawson, and Michael A. Mandell. Topological cyclic homology via the norm. *Doc. Math.*, 23:2101–2163, 2018.

[ACH22] Gabriel Angelini-Knoll, Dominic Leon Culver, and Eva Höning. Topological Hochschild homology of truncated Brown-Peterson spectra II. *In progress*, 2022.

[Ada66] J. F. Adams. On the groups $J(X)$. IV. *Topology*, 5:21–71, 1966.

[AGH21] Gabriel Angelini-Knoll, Teena Gerhardt, and Michael Hill. Real topological Hochschild homology via the norm and Real Witt vectors. *arXiv e-prints*, page arXiv:2111.06970, November 2021.

[AHL10] Vigleik Angeltveit, Michael A. Hill, and Tyler Lawson. Topological Hochschild homology of ℓ and *ko*. *Amer. J. Math.*, 132(2):297–330, 2010.

[AHW22] Gabriel Angelini-Knoll, Jeremy Hahn, and Dylan Wilson. Syntomic cohomology and K-theory of Morava K-theory. *In progress*, 2022.

[AK17] Gabriel J. Angelini-Knoll. *Periodicity in Iterated Algebraic K-Theory of Finite Fields*. ProQuest LLC, Ann Arbor, MI, 2017. Thesis (Ph.D.)–Wayne State University.

[AK18] Gabriel Angelini-Knoll. Detecting β elements in iterated algebraic K-theory of finite fields. *arXiv e-prints*, page arXiv:1810.10088, October 2018.

[AK21] Gabriel Angelini-Knoll. On topological Hochschild homology of the $K(1)$-local sphere. *Journal of Topology*, 14(1):258–290, 2021.

[AK22] Gabriel Angelini-Knoll. Complex orientations and TP of complete DVRS. *Accepted in Homol. Homotopy Appl.*, 2022.

[AMP22] Gabriel Angelini-Knoll, Mona Merling, and Maximilien Péroux. Homology of twisted g-rings. *In progress*, 2022.

[AQ19] Gabriel Angelini-Knoll and J. D. Quigley. Chromatic Complexity of the Algebraic K-theory of $y(n)$. *arXiv e-prints*, page arXiv:1908.09164, August 2019.

[AQ21] Gabriel Angelini-Knoll and J. D. Quigley. The Segal conjecture for topological Hochschild homology of Ravenel spectra. *Journal of Homotopy and Related Structures*, 2021.

[AQ22] Gabriel Angelini-Knoll and J.D. Quigley. Equivariant motivic filtrations. *In progress*, 2022.

[AR02] Christian Ausoni and John Rognes. Algebraic K-theory of topological K-theory. *Acta Math.*, 188(1):1–39, 2002.

[AR08] Christian Ausoni and John Rognes. *The chromatic red-shift in algebraic K-theory*, volume 40 of *Monogr. Enseign. Math.* L'Enseignement Mathématique, Geneva, 2008.

[AR12] Christian Ausoni and John Rognes. Algebraic K-theory of the first Morava K-theory. *J. Eur. Math. Soc. (JEMS)*, 14(4):1041–1079, 2012.

[AS18] Gabe Angelini-Knoll and Andrew Salch. A May-type spectral sequence for higher topological Hochschild homology. *Algebr. Geom. Topol.*, 18(5):2593–2660, 2018. agt.2018.18.2593.

[AS20] Gabriel Angelini-Knoll and Andrew Salch. Commuting unbounded homotopy limits with Morava K-theory. *Submitted.*, 2020. arXiv:2003.03510.

[Beh06] Mark Behrens. A modular description of the $K(2)$-local sphere at the prime 3. *Topology*, 45(2):343–402, 2006.

[BGHL18] A. J. Blumberg, T. Gerhardt, M. A. Hill, and T. Lawson. The Witt vectors for Green functors. *ArXiv e-prints*, March 2018.

[BHM93] M. Bökstedt, W. C. Hsiang, and I. Madsen. The cyclotomic trace and algebraic K-theory of spaces. *Invent. Math.*, 111(3):465–539, 1993.

[BHS20] Robert Burklund, Jeremy Hahn, and Andrew Senger. Galois reconstruction of Artin-Tate $\mathbb{R}$-motivic spectra. *arXiv e-prints*, page arXiv:2010.10325, October 2020.

[BL09] Mark Behrens and Gerd Laures. β-family congruences and the f-invariant. In *New topological contexts for Galois theory and algebraic geometry (BIRS 2008)*, volume 16 of *Geom. Topol. Monogr.*, pages 9–29. Geom. Topol. Publ., Coventry, 2009.

[BL22] Bhargav Bhatt and Jacob Lurie. Absolute prismatic cohomology. *arXiv e-prints*, page arXiv:2201.06120, January 2022.

[BM94] M. Bökstedt and I. Madsen. Topological cyclic homology of the integers. *Astérisque*, (226):7–8, 57–143, 1994. K-theory (Strasbourg, 1992).

[BMS19] Bhargav Bhatt, Matthew Morrow, and Peter Scholze. Topological Hochschild homology and integral p-adic Hodge theory. *Publ. Math. Inst. Hautes Études Sci.*, 129:199–310, 2019.

[Bök87] M. Bökstedt. The topological Hochschild homology of $\mathbb{Z}$ and of $\mathbb{Z}/p\mathbb{Z}$. preprint, 1987.

[BS19] Bhargav Bhatt and Peter Scholze. Prisms and Prismatic Cohomology. *arXiv e-prints*, page arXiv:1905.08229, May 2019.

[DGM13] Bjørn Ian Dundas, Thomas G. Goodwillie, and Randy McCarthy. *The local structure of algebraic K-theory*, volume 18 of *Algebra and Applications*. Springer-Verlag London, Ltd., London, 2013.

[DK14] Tobias Dyckerhoff and Mikhail Kapranov. Crossed simplicial groups and structured surfaces. *arXiv e-prints*, page arXiv:1403.5799, March 2014.

[DMPP17] E. Dotto, K. Moi, I. Patchkoria, and S. Precht Reeh. Real topological Hochschild homology. *ArXiv e-prints*, November 2017.

[GCH21] Angelini-Knoll, Gabriel, Dominic Leon Culver, and Eva Höning. Topological Hochschild homology of truncated Brown-Peterson spectra I. *arXiv e-prints*, page arXiv:2106.06785, June 2021.

[GIKR18] Bogdan Gheorghe, Daniel C. Isaksen, Achim Krause, and Nicolas Ricka. C-motivic modular forms. 10 2018.

[Hes97] Lars Hesselholt. Witt vectors of non-commutative rings and topological cyclic homology. *Acta Math.*, 178(1):109–141, 1997.

[HHR16] M. A. Hill, M. J. Hopkins, and D. C. Ravenel. On the nonexistence of elements of Kervaire invariant one. *Ann. of Math. (2)*, 184(1):1–262, 2016.

[HM03] Lars Hesselholt and Ib Madsen. On the K-theory of local fields. *Ann. of Math. (2)*, 158(1):1–113, 2003.

[HN19] Lars Hesselholt and Thomas Nikolaus. *Topological cyclic homology.* CRC Press, 2019.

[HNS22] Yonaton Harpaz, Thomas Nikolaus, and J. Shah. The real cyclotomic trace map. *In progress*, 2022.

[HRW22] Jeremy Hahn, Arpon Raksit, and Dylan Wilson. A motivic filtration on the topological cyclic homology of commutative ring spectra. *arXiv e-prints*, page arXiv:2206.11208, June 2022.

[HSW20] Jeremy Hahn, Andrew Senger, and Dylan Wilson. Odd primary analogs of Real orientations. *arXiv e-prints*, page arXiv:2009.12716, September 2020.

[HW20] Jeremy Hahn and Dylan Wilson. Redshift and multiplication for truncated Brown-Peterson spectra. *arXiv e-prints*, page arXiv:2012.00864, December 2020.

[IWX20] Daniel C. Isaksen, Guozhen Wang, and Zhouli Xu. Stable homotopy groups of spheres. *Proceedings of the National Academy of Sciences*, 117(40):24757–24763, 2020.

[Lau99] Gerd Laures. The topological q-expansion principle. *Topology*, 38(2):387–425, 1999.

[Lic73] Stephen Lichtenbaum. Values of zeta-functions, étale cohomology, and algebraic K-theory. pages 489–501. Lecture Notes in Math., Vol. 342, 1973.

[MR99] Mark Mahowald and Charles Rezk. Brown-Comenetz duality and the Adams spectral sequence. *Amer. J. Math.*, 121(6):1153–1177, 1999.

[NS18] Thomas Nikolaus and Peter Scholze. On topological cyclic homology. *Acta Math.*, 221(2):203–409, 2018.

[Pst18] Piotr Pstragowski. Synthetic spectra and the cellular motivic category. 03 2018.

[Qui72] Daniel Quillen. On the cohomology and K-theory of the general linear groups over a finite field. *Ann. of Math. (2)*, 96:552–586, 1972.

[Rog00] John Rognes. Algebraic k-theory of

finitely presented ring spectra, 2000. Talk at Oberwolfach.

[Wal84] Friedhelm Waldhausen. Algebraic K-theory of spaces, localization, and the chromatic filtration of stable homotopy. In *Algebraic topology, Aarhus 1982 (Aarhus, 1982)*, volume 1051 of *Lecture Notes in Math.*, pages 173–195. Springer, Berlin, 1984.